

Week 1 Module

1. Syntax

- set of rules that describes the structure and format of a programming language.
- it defines how commands and instructions are written in a way that the computer can understand and execute.
- each programming language has its own syntax, which developers must follow to create valid, error-free functional programs.

Syntax for C :-

```
#include <stdio.h>
```

```
int main()  
{  
    //code here  
    return 0;  
}
```

Example :-

```
#include <stdio.h>
```

```
int main()  
{  
    printf("Hello World!");  
  
    return 0;  
}
```

Output :-

Hello World!

2. Conditions and Loops

Conditional statements execute specific block of code based on whether a condition is true or false.

→ In C, 0 is false and any non-zero value is true.

if statement:- executes a block of code only if the specified condition is true.

Eg:-

```
#include <stdio.h>
int main()
{
    int age;
    printf("enter age : ");
    scanf("%d", &age);

    if (age >= 18)
    {
        printf("adult");
    }

    return 0;
}
```

Input:-

enter age : 18

Output:-

adult

Syntax:-

```
if (condition)
{
    // code here
}
```

if - else statement:- Provides an alternative block of code only if the specified condition is true. that means if the initial if condition evaluates to false.

Syntax:-

```
if (condition)
{
    // code if condition is true
} else {
    // code if condition is false
}
```

Eg:- `#include <stdio.h>`
`int main()`
`{`
`int n;`
`printf("enter a number: ");`
`scanf("%d", &n);`

<u>Input:-</u> enter a number: 4 <u>Output:-</u> even
--

`if (n % 2 == 0)`
`{`
`printf("even");`
`} else {`
`printf("odd");`
`}`
`return 0;`
`}`

else-if ladder:- used to check multiple successive conditions.
The program executes the first block where the condition is true and skips the rest.

syntax:-

```

if (condition1)
{
    //code if condition1 is true
} else if (condition2) {
    //code if condition2 is true
} else {
    //code if conditions are false
}

```

Eg:- #include <stdio.h>

int main()

{

int score = 75;

if (score >= 90)

{

printf("Grade : A\n");

} else if (score >= 75) {

printf("Grade : B\n");

} else {

printf("Grade : C\n");

}

return 0;

}

Output:-

Grade: B

switch statement:- Alternative to a long if-else ladder. It tests a single variable against a list of predefined values called cases and jumps directly to the matching one. It typically requires a break keyword to stop execution from bleeding into the next case.

Syntax:-

switch(expression)

{

case value1:

// code to execute

break;

case value2:

// code to execute

break;

default:

// code to execute

}

Eg:- #include <stdio.h>

int main()

{

int day = 2;

switch(day)

{

case 1:

printf("Monday\n");

break;

case 2:

printf("Tuesday\n");

break;

default:

printf("Invalid day\n");

}

Output:-

Tuesday

Loops

are used to repeat a block of code multiple times as long as a specified condition remains true.

for Loop:- used when we know exactly how many times the loop should run.

→ it groups three things in one line: initialization (counter start), condition (when to stop), and increment/decrement (how the counter changes)

Syntax:-

```
for(initialization; condition; increment/decrement)
{
```

// code here

```
}
```

Eg:- #include <stdio.h>

int main()

{

for (int i = 1; i <= 5; i++)

{
printf("%d", i);
}

return 0;

}

Output:-

1 2 3 4 5

while loop:- → "entry-controlled" loop that checks the condition before executing the loop body.

→ it is used when the exact number of iteration is unknown beforehand.

Syntax:- initialization;

while (condition)

{

// code here

increment/decrement;

}

Eg:- #include <stdio.h>

int main()

{

int i = 1;

while (i <= 5)

{

printf("%d", i);

i++;

}

return 0;

}

Output:-

1 2 3 4 5

do-while loop :- $\rightarrow$ "exit-controlled" loop that executes the code block at least once before checking the condition at end.

Syntax :- initialization ;
do
{
 // code here
 increment / decrement ;
} while (condition) ;

Eg :-
#include <stdio.h>
int main()
{
 int i = 10;
 do {
 printf("This runs at least once !\n");
 i++;
 } while (i < 5);

 return 0;
}

Output :-
This runs at least
once !

3. Programming Problems

1. Square

```
*****  
*****  
*****  
*****  
*****
```

```
#include <stdio.h>  
  
int main()  
{  
  
    int n = 5;  
  
    // for rows  
    for (int i = 0; i < n; i++)  
    {  
        // for columns  
        for (int j = 0; j < n; j++)  
        {  
            printf("* ");  
        }  
        printf("\n");  
    }  
  
    return 0;  
}
```



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main.c	Run	Output
<pre>1 #include<stdio.h> 2 int main() 3 { 4 int n = 5; 5 //for rows 6 for(int i = 0; i < n; i++) 7 { 8 //for columns 9 for(int j = 0; j< n; j++) 10 { 11 printf("* "); 12 } 13 printf("\n"); 14 } 15 return 0; 16 }</pre>	<pre>* === Code Exe</pre>	

2. Hollow square

* * * * *

* * * * *

* * * * *

* * * * *

* * * * *

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n = 5;
```

```
    // for first row
```

```
    for (int j = 0; j < n; j++)
```

```
    {
```

```
        printf("*");
```

```
    }
```

```
    printf("\n");
```

```
    // for mid rows
```

```
    for (int i = 1; i < n - 1; i++)
```

```
    {
```

Method 1

```

for(int j=0; j<n; j++)
{
    if(j==0 || j==n-1)
        printf("*");
    else {
        printf(" ");
    }
}
printf("\n");
}

```

// for last row

```

for(int j=0; j<n; j++)
{
    printf("*");
}
return 0;
}

```



main.c



Run

Output

```
1 #include<stdio.h>
2 int main()
3 {
4     int n = 5;
5     //printing first row
6     for(int j = 0; j < n; j++)
7     {
8         printf("* ");
9     }
10    printf("\n");
11
12    //printing middle row
13    for(int i = 1; i < n - 1; i++)
14    {
15        for(int j = 0; j < n; j++)
16        {
17            if(j==0 || j == n - 1)
18            {
19                printf("* ");
20            } else {
21                printf(" ");
22            }
23        }
24        printf("\n");
25    }
26
27    //printing last row
28    for(int j = 0; j < n; j++)
29    {
30        printf("* ");
31    }
32    printf("\n");
33    return 0;
34 }
```

```
* * * * *
*       *
*       *
*       *
* * * * *
```

=== Code Exec

Method-2

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n=5
```

```
    for(int i=0; i<n; i++)
```

```
    {
```

```
        for(int j=0; j<n; j++)
```

```
        {
```

```
            if (i==0 || i==n-1 || j==0 || j==n-1)
```

```
            {
```

```
                printf("*");
```

```
            }
```

```
            else {
```

```
                printf(" ");
```

}

}

printf("\n");
}

return 0;
}



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main.c



Run

Output

```
1 #include<stdio.h>
2 int main()
3 {
4     int n = 5;
5     for(int i = 1; i <= n; i++)
6     {
7         for(int j = 1; j <= n; j++)
8         {
9             if(i==1 || i==n || j==1 || j==n)
10            {
11                printf("* ");
12            } else {
13                printf("  ");
14            }
15        }
16        printf("\n");
17    }
18    return 0;
19 }
```

```
* * * * *
*       *
*       *
*       *
* * * * *
```

=== Code Executed

3 Triangle

*
**


```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n=5;
```

```
    for(int i=0; i<n; i++)
```

```
    {
```

```
        for(int j=0; j<=i; j++)
```

```
        {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```



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main.c



Run

Output

```
1 #include<stdio.h>
2 int main()
3 {
4     int n = 5;
5     //for rows
6     for(int i = 0; i < n; i++)
7     {
8         //for columns
9         for(int j = 0; j <= i; j++)
10        {
11            printf("* ");
12        }
13        printf("\n");
14    }
15    return 0;
16 }
```

```
*
* *
* * *
* * * *
* * * * *
```

=== Code Ex

4. Inverted Triangle

* * * * *

* * * *

* * *

* *

*

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n = 5;
```

```
    for(int i = 0; i < n; i++) {
```

```
        for(int j = 0; j < n - i; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```



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3



main.c



Run

Output

```
1 #include<stdio.h>
2 int main()
3 {
4     int n = 5;
5     //for rows
6     for(int i = 0; i < n; i++)
7     {
8         //for columns
9         for(int j = 0; j <= n - i - 1; j++)
10        {
11            printf("* ");
12        }
13        printf("\n");
14    }
15    return 0;
16 }
```

```
* * * * *
* * * *
* * *
* *
*
```

=== Code Ex

5. Pyramid

```
*
***
*****
*****
*****
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n=5;
```

```
for(int i=0; i<n; i++)
```

```
{
```

```
// spaces
```

```
for(int j=0; j<2*i+1; j++)
```

```
{
```

```
printf(" ");
```

```
}
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n=5;
```

```
for(int i=0; i<n; i++)
```

```
{
```

```
// spaces
```

```
for(int j=0; j<n-i-1; j++)
```

```
{
```

```
printf(" ");
```

```
}
```

```
// stars
```

```
for(int j=0; j<2*i+1; j++)
```

```
{
```

```
printf("*");
```

```
}
```

```
printf("\n");
```

```
}
```

```
return 0;
```



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3



main.c



Run

Output

```
1 #include<stdio.h>
2 int main()
3 {
4     int n = 5;
5     for(int i = 0; i < n; i++)
6     {
7         //for spaces
8         for(int j = 0; j < n - i - 1; j++)
9         {
10             printf(" ");
11         }
12
13         //for stars
14         for(int j = 0; j < 2*i + 1; j++)
15         {
16             printf("* ");
17         }
18         printf("\n");
19     }
20     return 0;
21 }
```

```
      *
    * * *
  * * * * *
* * * * * * *
* * * * * * * * *
```

=== Code Execution Success

6. Inverted Pyramid

**

*

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n=5;
```

```
    for(int i=0; i<n; i++)
```

```
    {
```

```
        // spaces
```

```
        for(int j=0; j<i; j++)
```

```
        {
```

```
            printf(" ");
```

```
        }
```

```
        // stars
```

```
        for(int j=0; j<2*(n-i)-1; j++)
```

```
        {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```



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4



main.c



Run

Output

```
1 #include<stdio.h>
2 int main()
3 {
4     int n = 5;
5     for(int i = 0; i < n; i++)
6     {
7         //for spaces
8         for(int j = 0; j < i; j++)
9         {
10             printf(" ");
11         }
12
13         //for stars
14         for(int j = 0; j < 2*(n - i) - 1; j++)
15         {
16             printf("*");
17         }
18         printf("\n");
19     }
20     return 0;
21 }
```

*

=== Code Ex

7. Half Diamond

```
*
**
***
****
*****
****
***
**
*
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n=5;
```

```
//printing first five lines (triangle)
```

```
for(int i=0; i<n; i++)
```

```
{
```

```
for(int j=0; j<=i; j++)
```

```
{
```

```
printf("*");
```

```
}
```

```
printf("\n");
```

```
}
```

```
//printing last four lines (inverted triangle)
```

```
for(int i=0; i<n-1; i++)
```

```
{
```

```
for(int j=0; j<n-i-1; j++)
```

```
{
```

```
printf("*");
```

```
}
```

```
printf("\n");
```

```
}
```

```
return 0;
```

```
}
```



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main.c



Run

Output

```
1 #include<stdio.h>
2 int main()
3 {
4     int n = 5;
5     //for printing the first five lines(triangle
6     )
7     for(int i = 0; i < n; i++)
8     {
9         for(int j = 0; j <= i; j++)
10            printf("*");
11        }
12        printf("\n");
13    }
14    //for printing the last four lines(inverted
15    triangle )
16    for(int i = 0; i < n - 1; i++)
17    {
18        for(int j = 0; j < n - i - 1; j++)
19            printf("*");
20        }
21        printf("\n");
22    }
23    return 0;
24 }
```

```
*
**
***
****
*****
****
***
**
*
```

=== Code

8. Diamond

```

      *
     ***
    *****
   *********
  ***********
 *****
  *****
   ***
    *
  
```

```

#include <stdio.h>

int main()
{
    int n=5;

    //printing upright pyramid
    for(int i=0; i<n; i++)
    {
        for(int j=0; j<n-i-1; j++)
        {
            printf(" ");
        }

        for(int j=0; j<2*i+1; j++)
        {
            printf("*");
        }

        printf("\n");
    }

    //printing inverted pyramid
    for(int i=0; i<n-1; i++)
    {
        for(int j=0; j<i+1; j++)
        {
            printf(" ");
        }

        for(int j=0; j<2*(n-2-i)+1; j++)
        {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
  
```



main.c



Run

Output

```
1 #include<stdio.h>
2 int main()
3 {
4     int n = 5;
5     //printing upright pyramid
6     for(int i = 0; i < n; i++)
7     {
8         for(int j = 0; j < n - i - 1; j++)
9         {
10             printf(" ");
11         }
12         for(int j = 0; j < 2*i + 1; j++)
13         {
14             printf("*");
15         }
16         printf("\n");
17     }
18     //for inverted pyramid
19     for(int i = 0; i < n - 1; i++)
20     {
21         for(int j = 0; j < i + 1; j++)
22         {
23             printf(" ");
24         }
25         for(int j = 0; j < 2*(n-2-i) + 1; j++)
26         {
27             printf("*");
28         }
29         printf("\n");
30     }
31     return 0;
32 }
```

```
*
***
*****
*****
*****
*****
***
*
```

=== Code Execut

1. Syntax

'C++'

```
#include <iostream>
using namespace std;
```

```
int main()
{
    // code here
    return 0;
}
```

Eg:- #include <iostream>
using namespace std;

```
int main()
{
    cout << "Hello World!";
    return 0;
}
```

2. Conditions and Loops

if-statement

syntax:- if(condition)
{
 // code
}

Eg:- #include <iostream>
using namespace std;
int main() {

```
    int speed = 90;  
    if (speed > 80)  
    {
```

```
        cout << "slow down" << endl;  
    }  
    return 0;  
}
```

Output:-
slow down

if-else statement:-

syntax:-

```
if (condition)
```

```
{  
    // code if true  
}  
else {  
    // code if false  
}
```

Eg:-

```
#include <iostream>  
using namespace std;
```

```
int main()  
{
```

```
    int age = 17;
```

```
    if (age >= 18)
```

```
    {  
        cout << "You can vote." << endl;
```

```
    } else {
```

```
        cout << "Too young to vote." << endl;
```

```
    }
```

```
    return 0;  
}
```

Output:-

Too young to vote.

else-if ladder:-

syntax:-

```
if (condition1) {
```

```
    // code
```

```
} else if (condition2) {
```

```
    // code
```

```
} else {
```

```
    // code if none match
```

```
}
```

Eg:- #include <iostream>
using namespace std;

int main()
{

int temperature = 32;
if (temperature > 30)
{

cout << "It is hot outside." << endl;

} else if (temperature > 15) {

cout << "It is warm." << endl;

} else {

cout << "It is cold." << endl;

}

return 0;

}

Output:-

It is hot outside.

switch statement:-

Syntax: switch (expression)
{

case value1:

//code

break;

case value2:

//code

break;

default:

//code if no match

}

Eg:-

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int rating = 2;
    switch (rating)
    {
```

code 1:

```
cout << "Poor Performance." << endl;
break;
```

code 2:

```
cout << "Average Performance." << endl;
break;
```

default:

```
cout << "Excellent Performance." << endl;
```

```
}
```

```
return 0;
```

```
}
```

Output:-

Average Performance.

for loop

Syntax:-

```
for (initialization; condition; increment/decrement)
{
```

```
// code to repeat
```

```
}
```

Eg:- `#include <iostream>`
`using namespace std;`

```
int main()
{
    cout << "Counting forward:";
    for(int i=1; i<=5; i++)
    {
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

Output:-
Counting forward: 1 2 3 4 5

while loop:-

Syntax:- `while(condition)`
`{`
`//code to repeat`
`}`

Eg:-

```
#include <iostream>
using namespace std;
int main()
{
```

```
    int count = 1;
    while (count <= 3)
    {
```

```
        cout << "Hello " << count << endl;
        count++;
    }
```

```
    return 0;
}
```

Output:-

Hello 1

Hello 2

Hello 3

do-while loop :-

syntax :-
do
{
 // code to repeat
} while (condition);

Eg :-
#include <iostream>
using namespace std;
int main()
{

 int x = 10;

 do

 {

 cout << "This prints even though x is not less
 than 5." << endl;

 x++;

 } while (x < 5);

 return 0;

}

Output :-

This prints even though x is
not less than 5.

3. Programming Problems

1. square

* * * * *

* * * * *

* * * * *

* * * * *

* * * * *

```
#include <iostream>
using namespace std;
int main()
```

```
{
```

```
    int n = 5;
```

```
    for(int i=0; i<n; i++)
    {
```

```
        for(int j=0; j<n; j++)
        {
```

```
            cout << "*";
```

```
        }
```

```
cout << "\n";
```

```
}
```

```
return 0;
```

```
}
```



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main.cpp



Run

Output

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n = 5;
6     //for rows
7     for (int i = 0; i < n; i++)
8     {
9         //for columns
10        for (int j = 0; j < n; j++)
11        {
12            cout << "*";
13        }
14        cout << "\n";
15    }
16    return 0;
17 }
18
```

```
*****
*****
*****
*****
*****
```

=== Code Execution Su

2. Hollow square

* * * * *

* * * * *

* * * * *

* * * * *

* * * * *

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n = 5;
```

```
    for(int i = 0; i < n; i++)
```

```
    {
```

```
        for(int j = 0; j < n; j++)
```

```
        {
```

```
            if(i == 0 || i == n - 1 || j == 0 || j == n - 1)
```

```
            {
```

```
                cout << "*";
```

```
            } else {
```

```
                cout << " ";
```

```
            }
```

```
        }
```

```
    } cout << "\n";
```

```
    return 0;
```

```
}
```



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6



```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n = 5;
6     for (int i = 0; i < n; i++) {
7         for (int j = 0; j < n; j++) {
8             if (i == 0 || i == n - 1 || j == 0
9                 || j == n - 1) {
10                cout << "*";
11            } else {
12                cout << " ";
13            }
14            cout << "\n";
15        }
16        return 0;
17    }
18 }
```

* *

* *

* *

=== Code Executed

3. Triangle

*

**

```
#include <iostream>
using namespace std;
```

```
int main()
```

```
{
```

```
    int n = 5;
```

```
    // rows
```

```
    for(int i = 0; i < n; i++)
```

```
    {
```

```
        // columns
```

```
        for(int j = 0; j <= i; j++)
```

```
        {
```

```
            cout << "*" << " ";
```

```
        }
```

```
        cout << "\n";
```

```
    }
```

```
    return 0;
```

```
}
```



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6



```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n = 5;
6     for (int i = 0; i < n; i++) {
7         for (int j = 0; j <= i; j++) {
8             cout << "*";
9         }
10        cout << "\n";
11    }
12    return 0;
13 }
14
```

```
*
**
***
****
*****
```

=== Code Execution Success

4 Inverted Triangle

* * * * *

* * * *

* * *

* *

*

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n = 5;
```

```
    for (int i = 0; i < n; i++) {
```

```
        for (int j = 0; j < n - i; j++) {
```

```
            cout << " * ";
```

```
        }
```

```
        cout << "\n";
```

```
    }
```

```
    return 0;
```

```
}
```



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7



```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n = 5;
6     for (int i = 0; i < n; i++) {
7         for (int j = 0; j < n - i; j++) {
8             cout << "*";
9         }
10        cout << "\n";
11    }
12    return 0;
13 }
14
```

```
*****
****
***
**
*
```

=== Code Execution Success

5. Pyramid

```
      *  
     ***  
    *****  
   *********  
  ***********
```

```
#include <iostream>  
using namespace std;
```

```
int main()  
{
```

```
    int n=5;
```

```
    for(int i=0; i<n; i++)  
    {
```

```
        //spaces
```

```
        for(int j=0; j<n-i-1; j++)  
        {
```

```
            cout<<" ";
```

```
        }
```

```
        //stars
```

```
        for(int j=0; j<2*i+1; j++)  
        {
```

```
            cout<<"*";
```

```
        }
```

```
        cout<<"\n";
```

```
    }
```

```
    return 0;
```

```
}
```



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8



main.cpp



Run

Output

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n = 5;
6     for (int i = 0; i < n; i++) {
7         // Print spaces
8         for (int j = 0; j < n - i - 1; j++) {
9             cout << " ";
10        }
11        // Print stars
12        for (int j = 0; j < (2 * i + 1); j++) {
13            cout << "*";
14        }
15        cout << "\n";
16    }
17    return 0;
18 }
19
```

```
*
***
*****
*****
*****
```

=== Code Executed

6. Inverted Pyramid

```
*****
*****
*****
*****
*****
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n = 5;
```

```
    for(int i=0; i<n; i++)
```

```
    {
```

```
        // spaces
```

```
        for(int j=0; j<i; j++)
```

```
        {
```

```
            cout << " ";
```

```
        }
```

```
        // stars
```

```
        for(int j=0; j<2*(n-i)-1; j++)
```

```
        {
```

```
            cout << "*";
```

```
        }
```

```
        cout << "\n";
```

```
    }
```

```
    return 0;
```

```
}
```



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main.cpp



Run

Output

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n = 5;
6     for (int i = 0; i < n; i++) {
7         // Print spaces
8         for (int j = 0; j < i; j++) {
9             cout << " ";
10        }
11        // Print stars
12        for (int j = 0; j < (2 * (n - i) - 1); j
            ++ ) {
13            cout << "*";
14        }
15        cout << "\n";
16    }
17    return 0;
18 }
19
```

*

=== Code Exec

7. Half Diamond

```
*  
**  
***  
****  
*****  
****  
***  
**  
*
```

```
#include <iostream>  
using namespace std;
```

```
int main()  
{
```

```
int n = 5;
```

```
// top half triangle (including middle row)
```

```
for(int i = 0; i < n; i++)  
{
```

```
for(int j = 0; j <= i; j++)  
{
```

```
cout << "*";
```

```
}
```

```
cout << "\n";
```

```
}
```

```
// bottom half triangle
```

```
for(int i = 0; i < n - 1; i++)  
{
```

```
for(int j = 0; j < n - i - 1; j++)  
{
```

```
cout << "*";
```

```
}
```

```
cout << "\n";
```

```
}
```

```
return 0;
```

```
}
```



main.cpp



Run

Output

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n = 5;
6
7     // 1. Top Half triangle (including the
        middle row)
8     for (int i = 0; i < n; i++) {
9         for (int j = 0; j <= i; j++) {
10             cout << "*";
11         }
12         cout << "\n";
13     }
14
15     // 2. Bottom Half triangle
16     for (int i = 0; i < n - 1; i++) {
17         for (int j = 0; j < n - 1 - i; j++) {
18             cout << "*";
19         }
20         cout << "\n";
21     }
22
23     return 0;
24 }
25
```

```
*
**
***
****
*****
*****
****
***
**
*
```

=== Code Executed



main.cpp



Run

Output

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n = 5;
6
7     // 1. Top Half pyramid
8     for (int i = 0; i < n; i++)
9     {
10         for (int j = 0; j < n - i - 1; j++)
11         {
12             cout << " ";
13         }
14         for (int j = 0; j < (2 * i + 1); j++)
15         {
16             cout << "*";
17         }
18         cout << "\n";
19     }
20
21     // 2. Bottom Half inverted pyramid
22     for (int i = 0; i < n - 1; i++)
23     {
24         //
25         for (int j = 0; j <= i; j++)
26         {
27             cout << " ";
28         }
29         for (int j = 0; j < (2 * (n - 1 - i) - 1); j++)
30         {
31             cout << "*";
32         }
33         cout << "\n";
34     }
35
36     return 0;
37 }
38
```

```
*
***
*****
*****
*****
*****
***
*
```

=== Code Execution Suc